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COURSE: CSA0409 - Operating System

$CO_2 - AT_2$

1. CPU Scheduling in a Real-Time Ticket Booking System:

a) ALGORITHM:

Priority Scheduling.

b) JUSTIFICATION:

P_1 (Payment) has highest priority, so it gets CPU first. This reduces waiting time for important transactions & improves system response during heavy load

c) EXECUTION ORDER:

$P_1 \rightarrow P_2 \rightarrow P_3$

High $\rightarrow$ Medium $\rightarrow$ Low priority.

d) DRAWBACKS:

Low-priority P_3 may suffer starvation if high-priority processes keep arriving.

SOLUTION:

Aging

2. Deadlock in a Banking Transaction Server.

a) RAG:

$A \rightarrow T_1 \rightarrow B \rightarrow T_2 \rightarrow C \rightarrow T_3 \rightarrow A$

b) DEADLOCKS

Yes, T_1 waits for B, T_2 waits for C, and T_3 waits for A. This forms a circular wait, so no transaction can proceed

c) TECHNIQUE:

Use deadlock prevention with fixed resource ordering

d) LOCK ORDER:

$A \to B \to C$

All transaction must request resources in this order, preventing circular wait.

3. MOBILE OS APP SWITCHING LAG:

a) CPU IDLE:

Many apps are waiting for I/O, so they are not ready to use the CPU.

b) 20ms QUANTUM:

It can hurt responsiveness because 20 background apps get CPU turns before the foreground app.

c) BETTER METHOD:

Use MLFQ / Priority Scheduling with a smaller quantum (5-10ms) for interactive apps

d) BENEFIT:

Foreground apps get CPU faster
↓
less waiting
↓
smoother app switching
↓
better throughput

4. Deadlock Risk in an Operating System Update

a) Deadlock :

Yes , it can occur.

b) Circular Wait :

$M_1 \rightarrow R_2 \rightarrow M_2 \rightarrow R_3 \rightarrow M_3 \rightarrow R_1 \rightarrow M_1$

Each Module holds one resource & waits for another.

c) Solution :

Use Banker's Algorithm,

Allocate a resource only if the system remains in a safe state

d) Priority :

Run independent / critical modules first and delay modules involved in resource conflicts

This reduces waiting and improves CPU utilization .

5. Cloud Server CPU Utilization Analysis :

a) GIVEN :

$p = 0.8$

$n = 7$

FORMULA :

$U = 1 - p^n$

SOL :

$U = 1 - (0.8)^7$

$U = 1 - 0.2097$

$U = 0.7903$

U = 79.03%

b) INTERPRETATION:

The server is reasonably well utilized.

c) VMs:

More VMs → better CPU utilization and throughput, but too many cause contention.

Fewer VMs → more CPU idle time.

d) OS STRATEGIES:

* Multiprogramming
* I/O - aware scheduling.